Question # 1: Multiple choice question. [15 x 1pt]

- Which of the following data structures allow random access
) Arrays b) Linked list c) Trees d) a & b

- Which of the following data structures is not a linear data structure.
Arrays b) Linked list c) Trees d) a & b

- In static stack the top of the stack is
First element in the array b) Second element in the array
Last element in the array d) determined by a variable.

- The postfix expression of the infix expression. A+B*(C+D)/F+D*E is:_____
) AB+CD+*F/D+E* b) ABCD+*F/+DE*+ c) A*B+CD/F*DE++ d) A+*BCD/F*DE++

- In a circular queue of size n, how do you determine if the queue is full?
) Front == Rear b) (Rear + 1) % n == Front c) Rear == n - 1 d) Rear == Front - 1

6- What is the primary advantage of a circular queue over a linear queue?
a) Dynamic resizing b) Constant-time searching
c) Efficient space utilization d) Simpler implementation

7- Which operation is expensive in an array when compared to linked lists?
a) Accessing elements b) Searching for elements c) Inserting elements in the middle d) Traversing

8- Which is a potential drawback of using linked lists over arrays?
a) Fixed size b) Sequential memory allocation
c) Higher memory overhead due to pointers d) Inefficient traversal

9- If the pointer P has the value of NULL what will happen if this statement is executed: cout<< p->info;
a) The output is 0 b) The output is Random value c) No output d) Invalid code

10- Which operation is not efficient in a singly linked list compared to a doubly linked list?
a) Insertion at the head b) Insertion at the tail c) Deletion from the head d) Deletion from the tail

11- In order to get the information stored in a binary search tree in the descending order, one should traverse it in which of the following order?
a) Left , root, right b) root, left, right c)right, root, left d) right, left, root

12- What happens if you insert elements in sorted order into a binary search tree?
a) The tree becomes balanced b) The height remains logarithmic
) The tree becomes a linked list d) a & b

3- Which traversal method is used in BSTs to delete all nodes in a tree safely?
Preorder b) Inorder c) Postorder d) Level order

- What will happen if a recursive function doesn't have a proper base case?
The function will terminate immediately
The function will continue running indefinitely, causing a stack overflow
he function will return an incorrect result d) The function will throw a compile-time error

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c) The recursive function itself

b) The condition that stops the recursion

d) The number of recursive calls

*** Write the correct answers of the Question # 1 in the table below

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15

Question # 2: Fill in the blanks. [7 x 1pts]

1- The result of the following postfix expressions: 2 5 5 + * 8 - 1 3 + / = _____

2- In the following preorder traversal of a binary tree nodes: E,C,K,A,H,B,G,D,F, the root node is _____

3- How many nodes are in a full binary tree of height equals to 5 ? _____

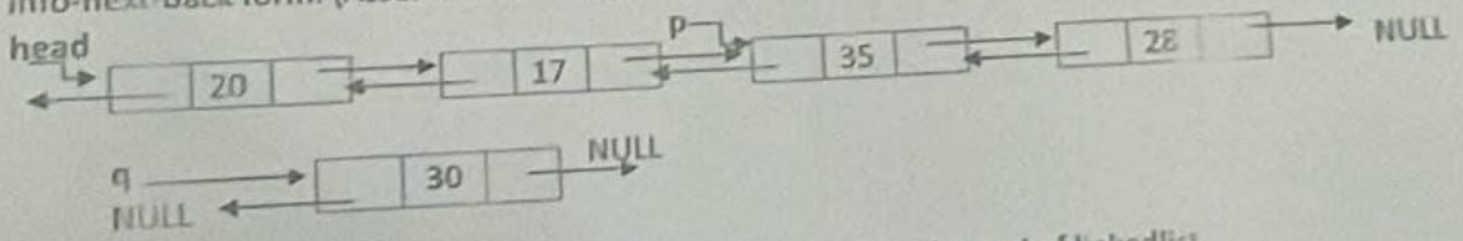
4- How many pointers are in doubly linked list must be updated to delete the last node? _____

5- How many pointers are in a binary tree must be updated to delete a node has 2 children ? _____

6- How many recursive calls are required to calculate the fifth element in Fibonacci numbers? _____

7- A circular queue is implemented using an array of size 10. The array index starts with 0, the front is 6, and the rear is 9. The insertion of the next element takes place at the array index _____

Question # 3: Consider the linked list shown in the Figure below. Assume that the nodes are in the u info-next-back form. (Assume that head,p and q are pointers of type node)[4pts]



a. Write two C++ statements to add the node with info 30 at the end of linkedlist

_____

_____

b. Give the value of the following expressions:

i. front-> next -> next -> back -> info _____

ii. p -> back -> next -> next -> info _____